Maximum Likelihood Estimators

Likelihood Function

Using the joint pdf or pmf of the sample X, the likelihood function is a function of θ, given the observed data $X = x$, defined as

$$L(\theta|x) = f(x|\theta)$$

non-identical

non independent

If the data is iid, then

univariate densities

$$f(x|\theta) = \cancel{\prod_{i=1}^{n}} f_i(x_i|\theta)$$

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Log-Likelihood Function

$$L(\theta) = \prod f(x_i)$$

$$\ln(L(\theta)) = \sum \ln f(x_i)$$

If $\ln\{L(\theta)\}$ is monotone of θ, then maximizing $\ell(\theta) = \ln\{L(\theta)\}$ will yield the maximum likelihood estimators.

Maximum log-Likelihood Estimator

What value of θ will yield the highest probability to create our observed data X

The maximum likelihood estimator are the estimates of θ that maximize $\ell(\theta)$.

The estimator is the best guess of the parameter.

Binomial Distribution

$x = 1 \quad p$

$x = 0 \quad 1-p$

Bernoulli(p) = Bin$(1, p)$

Let $X_1, \ldots, X_n \overset{iid}{\sim} \cancel{\text{Bin}(n, p)}$

$$f(x) = p^x (1-p)^{1-x}$$

$$L(p|x) = \prod_{i=1}^{n} p^{x_i} (1-p)^{1-x_i}$$

$$\frac{dL}{dp} = x_1 p^{x_1-1}\left[\underline{(1-p)^{1-x_1}} \prod_{i=2}^{n} p^{x_i}(1-p)^{x_i}\right] +$$

$$p^{x_1}\left[(1-x_1)(1-p)^{1-x_1-1} \underline{\prod_{i=2}^{n} p^{x_i}(1-p)^{x_i}}\right] +$$

$$\left[x_2 p^{x_2-1} \quad . \quad . \quad .\right.$$

$$\ell(\theta) = \ln\left[L(\theta)\right]$$

$$= \ln\left[\prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i}\right]$$

$$= \sum_{i=1}^{n} \ln\left[p^{x_i}(1-p)^{1-x_i}\right]$$

$$= \sum_{i=1}^{n}\left[x_i \ln p + (1-x_i)\ln(1-p)\right]$$

$$\frac{d\,\ell(p)}{dp} = \frac{d}{dp}\sum\left[x_i \ln p + (1-x_i)\ln(1-p)\right]$$

$$= \sum \frac{d}{dp}\left[x_i \ln p + (1-x_i)\ln(1-p)\right]$$

$$= \sum \left[\frac{x_i}{p} + \frac{1-x_i}{1-p}(-1)\right]$$

$$= \frac{1}{p}\sum x_i - \frac{1}{1-p}\sum(1-x_i)$$

$$\frac{\sum x_i}{p} - \frac{n-\sum x_i}{1-p} = 0$$

$$\frac{\sum x_i}{p} = \frac{n - \sum x_i}{1-p}$$

$$\sum x_i (1-p) = p(n - \sum x_i)$$

$$\sum x_i - \cancel{p \sum x_i} = pn - \cancel{p \sum x_i}$$

$$\sum x_i = pn$$

$$p = \frac{\sum x_i}{n} = \bar{x}$$

$$\hat{p} = \bar{x}$$

$$\frac{\Sigma x_i}{p} - \frac{n - \Sigma x_i}{1-p}$$

$$\frac{n\bar{x}}{p} - \frac{n - n\bar{x}}{1-p}$$

$$-\frac{n\bar{x}}{p^2} - \frac{(n - n\bar{x})}{(1-p)^2}$$

$$-\frac{n\bar{x}}{\bar{x}^2} - \frac{n(1-\bar{x})}{(1-\bar{x})^2}$$

$$\frac{-n}{\bar{x}} - \frac{n}{1-\bar{x}}$$

$$-n\left[\frac{1-\bar{x}+\bar{x}}{\bar{x}(1-\bar{x})}\right] = -n\,\frac{1}{\bar{x}(1-\bar{x})}$$

$X_1, \ldots, X_n \overset{iid}{\sim} Pois(\lambda)$

$$f_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$ Find MLE of λ

1. Likelihood
2. $\ell(x)$
3. Derivative
4. set 0 and solve.

$$L(\lambda|x) = \prod_{i=1}^{n} \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$\ell(\lambda) = \sum_{i=1}^{n} \ln\left[\frac{e^{-\lambda} \lambda^{x_i}}{x_i!}\right]$$

$$\ell(\lambda) = \sum \cancel{\ln(e^{-\lambda})}^{-\lambda} + x_i \ln(\lambda) - \ln(x_i!)$$

$$= \sum -\lambda + \ln(\lambda) \sum x_i - \sum \ln(x_i!)$$

$$= -n\lambda + \ln(\lambda) n\bar{x} - \sum \ln(x_i!)$$

$$\frac{d\ell}{d\lambda} = -n + \frac{n\bar{x}}{\lambda} = 0$$

$$\frac{n\bar{x}}{\lambda} = n$$

$$n\bar{x} = \lambda n$$

$$\underline{\bar{x}} = \hat{\lambda}$$

$X \sim Pois(\lambda)$

$x = 0, 1, 2, \ldots$

$\bar{x} > 0$

$$\frac{-n\bar{x}}{\lambda^2} = \frac{-n}{\bar{x}}$$

Normal Distribution

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$

$$L(\mu, \sigma^2 | x) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

$$\ell(\mu, \sigma^2 | x) = \sum_{i=1}^{n} \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{(x_i-\mu)^2}{2\sigma^2}$$

$$= \sum \left[-\frac{\ln(2\pi\sigma^2)}{2} - \frac{(x_i-\mu)^2}{2\sigma^2} \right]$$

$$= -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i-\mu)^2$$

$$\frac{d\ell(\mu,\sigma^2)}{d\mu} = -\frac{1}{2\sigma^2} \sum 2(x_i-\mu)(-1)$$

$$= \frac{\cancel{2}}{\cancel{2}\sigma^2} \sum (x_i-\mu) = 0$$

$$\sum (x_i-\mu) = 0$$

$$\sum x_i - n\mu = 0$$

$$\sum x_i = n\mu$$

$$\bar{x} = \hat{\mu} \qquad \frac{1}{y} \to -\frac{1}{y^2}$$

$$= -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i-\mu)^2$$

$$\frac{d\ell}{d\sigma^2} = -\frac{n}{2}\frac{1}{\cancel{2\pi}\sigma^2}\cancel{2\pi} + \frac{1}{2(\sigma^2)^2}\sum(x_i-\mu)^2$$

$$-\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum(x_i-\mu)^2 = 0$$

$$\frac{n}{2\sigma^2} = \frac{1}{2(\sigma^2)^2}\sum(x_i-\mu)^2$$

$$\frac{\cancel{2}(\sigma^2)^2 n}{\cancel{2}\sigma^2} = \sum(x_i-\mu)^2$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum(x_i-\mu)^2$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

Exponential Distribution

Let $X_1, \dots, X_n \overset{iid}{\sim} Exp(\lambda)$